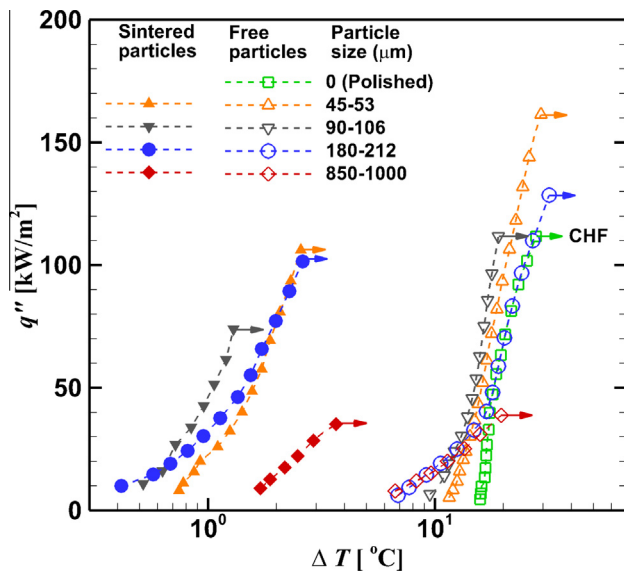




from the surface (and particle sizes), entrainment number of active nucleation sites at a given time instant. However, the significantly higher heat transfer coefficient compared to free particles. Nonetheless, the trends observed for the free-particle and sintered-coating are similar. Forming sintered coating with the same 90–106 μm particle size provides a 95% decrease in heat transfer that is 33% lower than the sintered coating has a polished surface. As the next step to implement, it is to study parametric trends in

Conflict of interest

None declared.

Acknowledgments

Fig. 8. Comparison of trends in boiling curves of area-averaged heat flux (q'') versus wall superheat (ΔT) for free and sintered particles with diameters ranging from 45 μm to 1000 μm.